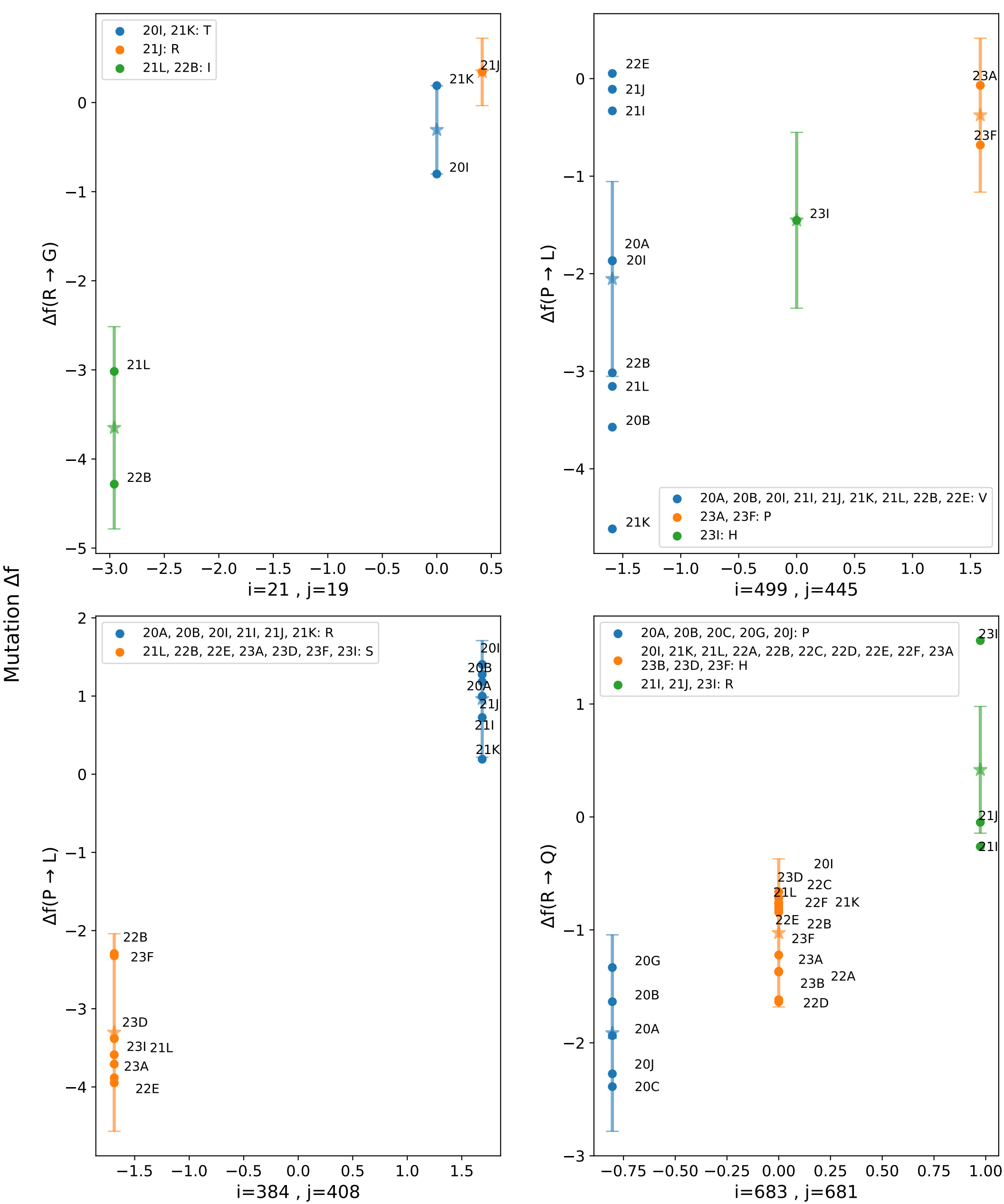
20I, 21K: T
21J: R
21L, 22B: I
21K
21J
20I
21L
22B
Δf(R → G)
i=21 , j=19
22E
21J
21I
23A
23F
23I
20A
20I
22B
21L
20B
21K
20A, 20B, 20I, 21I, 21J, 21K, 21L, 22B, 22E: V
23A, 23F: P
23I: H
Δf(P → L)
i=499 , j=445
20A, 20B, 20I, 21I, 21J, 21K: R
21L, 22B, 22E, 23A, 23D, 23F, 23I: S
20I
20B
20A
21J
21I
21K
22B
23F
23D
23I
21L
23A
22E
Δf(P → L)
i=384 , j=408
20A, 20B, 20C, 20G, 20J: P
20I, 21K, 21L, 22A, 22B, 22C, 22D, 22E, 22F, 23A
23B, 23D, 23F: H
21I, 21J, 23I: R
23I
21J
21I
20I
23D
22C
21L
22F
21K
22E
22B
23F
23A
23B
22A
22D
20G
20B
20A
20J
20C
Δf(R → Q)
i=683 , j=681
Mutation Δf
Coupling J